Substituting,

$$\lambda = \frac{\sin 30^\circ}{4\times10^5\times2} = \frac{0.5}{4\times10^5\times2}$$
$$= \frac{1}{2\times4\times10^5\times2} = \frac{1}{16\times10^5}$$

$$\lambda = 6250\times10^{-10}\,\text{m} = 6250\ \text{Å}$$

EXAMPLE 7.13

A monochromatic light of wavelength of 500 nm strikes a grating and produces fourth order maximum at an angle of 30°. Find the number of slits per centimeter.

Solution

λ = 500 nm = 500×10^{-9} m; m = 4;

θ = 30°; number of lines per cm = ?

Equation for diffraction maximum for grating is, sin $\theta = Nm\lambda$

Rewriting, $N = \frac{\sin\theta}{m\lambda}$

Substituting,

$$N = \frac{0.5}{4\times500\times10^{-9}}$$
$$= 2.5\times10^5\ \text{m}^{-1}$$
$$= 2.5\times10^3\ \text{cm}^{-1}$$

7.4.7 Experiment to determine the wavelength of monochromatic light

The wavelength of a spectral line can be very accurately determined with the help of a plane transmission grating. For that we need to use an instrument called spectrometer (Refer 7.6.6). After preliminary adjustments, the slit of collimator is illuminated by a monochromatic light, whose wavelength is to be determined. The telescope is brought in line with collimator to view the image of the slit. The given grating is then mounted on the prism table with its plane perpendicular to the incident beam of light coming from the collimator. The telescope is turned to one side until the first order diffraction image of the slit is seen. The reading of the position of the telescope is noted.

Similarly, the first order diffraction image on the other side is captured and the reading is noted. The difference between two readings gives 2θ. Half of its value gives θ. The angle for first order maximum is shown in Figure 7.21. The wavelength of light is calculated from the equation,

DO YOU KNOW?

A compact disc (CD) always appears colourful. On the read/writable side which looks shining, there are many narrow circular tracks with widths comparable to the wavelength of visible light. Hence, the diffraction takes place after the reflection of incident white light to give colourful appearance. The tracks act as reflecting grating.



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